

Computational Fluid Dynamics - 4/7/2024

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We want to simulate the evolution of a contaminant in the rectangular pool Ω sketched in Figure 1, filled with an incompressible Newtonian fluid. At the initial time $t = 0$ the fluid is at rest and the contaminant is uniformly distributed with concentration $c = 1$. The pool is cleaned by imposing an inflow of clean fluid (with $c = 0$) at the boundary Γ_{in} , while the contaminated fluid leaves the domain through the outflow Γ_{out} . The problem is governed by the following unsteady incompressible Navier–Stokes equations coupled with a time-dependent advection-diffusion equation for the concentration:

$$\begin{cases} \frac{\partial \mathbf{u}}{\partial t} - \operatorname{div}(\nu(\nabla \mathbf{u} + \nabla^T \mathbf{u})) + (\mathbf{u} \cdot \nabla) \mathbf{u} + \nabla p = 0, & \text{in } \Omega, t \in (0, t_f), \\ \operatorname{div} \mathbf{u} = 0, & \text{in } \Omega, t \in (0, t_f), \\ \frac{\partial c}{\partial t} - k \Delta c + \mathbf{u} \cdot \nabla c = 0, & \text{in } \Omega, t \in (0, t_f), \end{cases} \quad (1)$$

where ν is the kinematic viscosity of the fluid and k is the diffusivity of the contaminant.

We consider the following boundary conditions:

- for the flow problem we impose a parabolic velocity profile $\mathbf{u} = (10y(0.2 - y), 0)$ on Γ_{in} , zero traction on Γ_{out} and no-slip conditions on $\Gamma_{\text{wall}} = \partial\Omega \setminus (\Gamma_{\text{in}} \cup \Gamma_{\text{out}})$;
- for the concentration equation we impose $c = 0$ on Γ_{in} and homogeneous Neumann conditions on the remaining boundaries.



Figure 1: Computational domain

- a) Introduce a suitable time-advancing scheme for the time-discretization of the proposed problem such that, at each time step, continuity and momentum equations are first solved using the incremental Chorin-Temam projection method and then the concentration equation is solved using the updated velocity field.
- b) Write the weak formulations of the problems to solve at each time iteration discussing the choice of functional spaces and the treatment of boundary conditions.
- c) Write a Firedrake notebook, based on the template `template040724.ipynb`, implementing the proposed scheme using $\mathbb{P}_2$ finite elements for velocity and $\mathbb{P}_1$ finite elements for pressure and concentration.
- d) Given $\nu = 0.01$ and $k = 0.0001$, compute the solution until the final time $t_f = 100$ with a time step $\Delta t = 5$. Then:

- to monitor the cleaning procedure, evaluate the mean concentration at each time step over the whole domain and compute the time required to lower the mean concentration below one half of the initial value;
 - save plots of the velocity and temperature fields at the final time (export them as files `SURNAME_NAME_velocity.png`, `SURNAME_NAME_temperature.png`)
- e) Repeat the exercise considering now the same problem with a flowrate at the inflow ten times larger and simulate the flow until $t_f = 10$ with a time step $\Delta t = 0.5$. How does the solver behave in this case? Can the results be improved?